

Data Science Capstone Project Report

SpaceX Falcon 9 First Stage Landing Prediction

[Final PDF](#)

GitHub Repository URL: <https://github.com/Fi75-hub/Applied-Data-Science-Capstone>

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Project goal: predict whether the Falcon 9 first stage will land successfully and explain the launch conditions behind the prediction.

This completed PDF presentation covers rubric items 1.1 through 1.15.

Metric	Value
Launch records	90
Prepared features	83
Landing success	67%
Best test accuracy	89%

1.1 and 1.2 - GitHub Repository URL and Completed PDF

Repository URL, completed notebooks, completed Python files, and final PDF

GitHub Repository URL: <https://github.com/Fi75-hub/Applied-Data-Science-Capstone>

The GitHub repository contains all completed notebooks and Python files for this capstone project.

Final presentation file submitted in PDF format: Data Science Capstone Project Report.pdf.

Completed repository item	File name
SpaceX API data collection notebook	spacex_data_collection_api.ipynb
Web scraping notebook	spacex_web_scraping.ipynb
Data wrangling notebook	spacex_data_wrangling.ipynb
EDA visualization notebook	spacex_eda_visualization.ipynb
EDA SQL notebook	spacex_eda_sql.ipynb
Folium map notebook	spacex_folium_launch_sites.ipynb
Plotly Dash Python script	spacex_dash_app.py
Machine-learning notebook	spacex_machine_learning.ipynb

Repository evidence: the URL is provided plainly, and the file checklist shows the completed notebooks and Python script included in the project.

1.3 - Executive Summary

Summary of methods used and key results

Rubric 1.3

The project combined collection, wrangling, visualization, interactive analytics, and classification to answer the landing prediction question.

Area	Methods used	Key result
Data collection	SpaceX API calls and BeautifulSoup web scraping	Launch records were collected and structured for analysis.
Preparation	Date conversion, missing-value handling, target encoding, one-hot encoding	The Class target and 83-feature matrix were prepared.
EDA and SQL	Scatter plots, bar charts, yearly trends, grouped SQL queries	Launch site, orbit, payload, and time patterns affected success.
Interactive analytics	Folium map and Plotly Dash dashboard	Maps and dashboard filters explained geography and payload/outcome patterns.
Modeling	Logistic Regression, SVM, Decision Tree, and KNN	Logistic Regression was selected with 89% test accuracy.

Executive summary key result: landing success improved over time and is best explained by launch site, orbit, payload mass, and flight history together.

1.4 - Introduction

Project background and problem statement

Rubric 1.4

- SpaceX lowers launch cost when the Falcon 9 first stage lands and can be reused.
- A competing company can estimate launch cost and bid risk by predicting first-stage landing success.
- Business problem: identify the launch conditions that influence first-stage landing success.
- Data science question: can launch site, orbit, payload mass, flight number, booster reuse history, and mission features predict a successful landing?
- Prediction type: binary classification where Class = 1 is successful landing and Class = 0 is unsuccessful landing.

1.5 - Data Collection Methodology - SpaceX API

Rubric 1.5

API calls using key phrases

- API source: <https://api.spacexdata.com/v4/launches/past>.
- Key phrases: GET launch records, normalize JSON, extract rockets, payloads, cores, and launchpads.
- Expanded fields: launch site, latitude, longitude, orbit, payload mass, booster version, grid fins, legs, reuse, and landing outcome.
- Output: dataset_part_1.csv was saved for wrangling, EDA, dashboarding, and classification.
- GitHub Repository URL: <https://github.com/Fi75-hub/Applied-Data-Science-Capstone>.

API flow: request response -> json normalize -> expand IDs -> clean fields -> save dataset.

1.6 - Data Collection Methodology - Web Scraping

Rubric 1.6

Scraping process using key phrases

- The web-scraping notebook used requests and BeautifulSoup to parse Falcon 9 launch table rows.
- Key phrases: get HTML, parse table, extract rows, clean cells, standardize dates, convert payload values, and map landing outcomes.
- Scraped fields included flight number, launch date, booster version, launch site, payload, orbit, customer, and outcome.
- The scraped data acted as a second data collection workflow and supported validation of the API data.
- GitHub Repository URL: <https://github.com/Fi75-hub/Applied-Data-Science-Capstone>.

Parser used: BeautifulSoup with html.parser. Notebook: spacex_web_scraping.ipynb.

1.7 - Data Wrangling Methodology

Data cleaning and processing

Rubric 1.7

- Converted launch dates into datetime and extracted launch year for trend analysis.
- Filled missing payload mass values so records were not removed unnecessarily.
- Converted landing outcome into the binary Class target for classification.
- One-hot encoded categorical features such as orbit, launch site, landing pad, and booster serial.
- Prepared dataset_part_2.csv for analysis and dataset_part_3.csv for model training.
- GitHub Repository URL: <https://github.com/Fi75-hub/Applied-Data-Science-Capstone>.

1.8 - EDA with Data Visualization Methodology

Rubric 1.8

Charts used, purpose, and repository URL

- Scatter plots compared flight number and payload mass against launch site and landing outcome.
- Bar charts compared landing success rates across orbit and launch site categories.
- Yearly trend charts showed launch volume and landing success rate changes over time.
- Purpose: identify patterns that explain landing success and guide feature selection for classification.
- GitHub Repository URL: <https://github.com/Fi75-hub/Applied-Data-Science-Capstone>.

1.9 - EDA with SQL Methodology

SQL queries performed and repository URL

Rubric 1.9

- Launch-site queries counted launches and ranked sites by success rate.
- Payload queries calculated minimum, average, and maximum payload mass by outcome.
- Success-rate queries compared landing performance across sites, orbits, and years.
- Time-analysis queries grouped records by launch year to show operational improvement.
- GitHub Repository URL: <https://github.com/Fi75-hub/Applied-Data-Science-Capstone>.

1.10 - Interactive Visual Analytics Methodology

Rubric 1.10

Folium map and Plotly Dash app

- Folium map summarized launch site markers, launch records, and proximity to coastline, highway, railway, and city locations.
- Plotly Dash app summarized success distribution and payload-versus-outcome patterns with interactive controls.
- Dashboard controls included a launch-site dropdown and payload range slider.
- The callback updated the success pie chart and payload scatter chart from the selected inputs.
- GitHub files: `spacex_folium_launch_sites.ipynb` and `spacex_dash_app.py`. GitHub Repository URL: <https://github.com/Fi75-hub/Applied-Data-Science-Capstone>.

1.15 - Predictive Analysis Classification Methodology

Rubric 1.15

Classification slide with target, features, and models

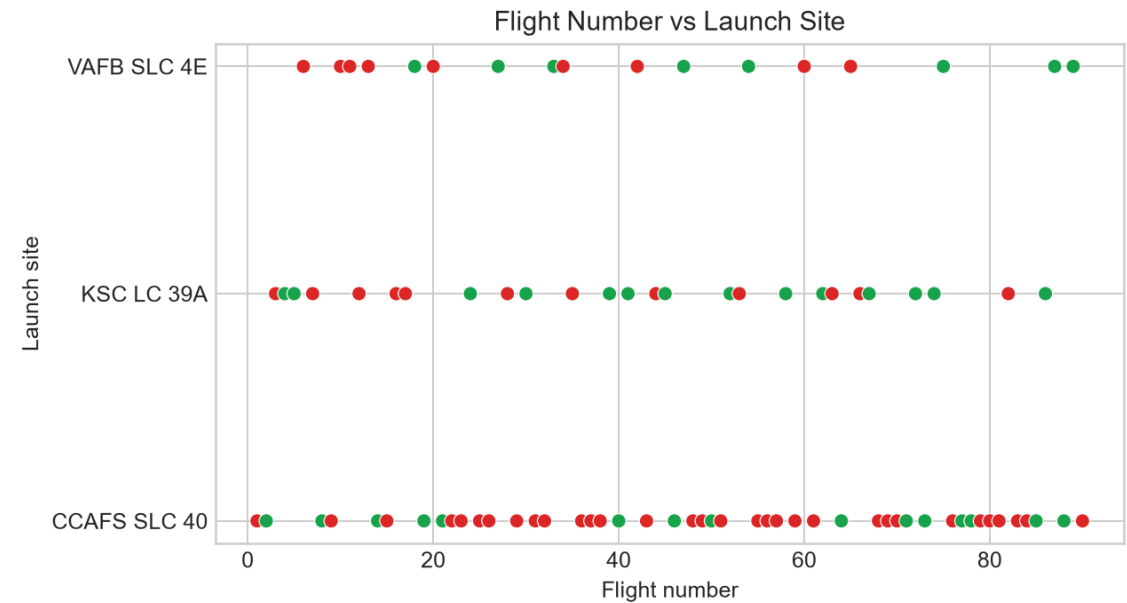
- Target variable: Class, where 1 indicates successful first-stage landing and 0 indicates unsuccessful landing.
- Features: payload mass, flight number, booster block, reuse count, flights, orbit, launch site, landing pad, grid fins, reused, and legs.
- Models tested: Logistic Regression, Support Vector Machine, Decision Tree, and K Nearest Neighbors.
- Evaluation method: held-out test accuracy and confusion-matrix behavior.
- GitHub Repository URL: <https://github.com/Fi75-hub/Applied-Data-Science-Capstone>.

1.11 - EDA with Visualization Results - Scatter Plot 1

Rubric 1.11

Flight number versus launch site

- Later flight numbers show more successful landings, which indicates operational improvement.
- Launch sites show different outcome patterns, making launch site an important explanatory feature.
- This slide satisfies the required scatter plot result for flight number and launch site.

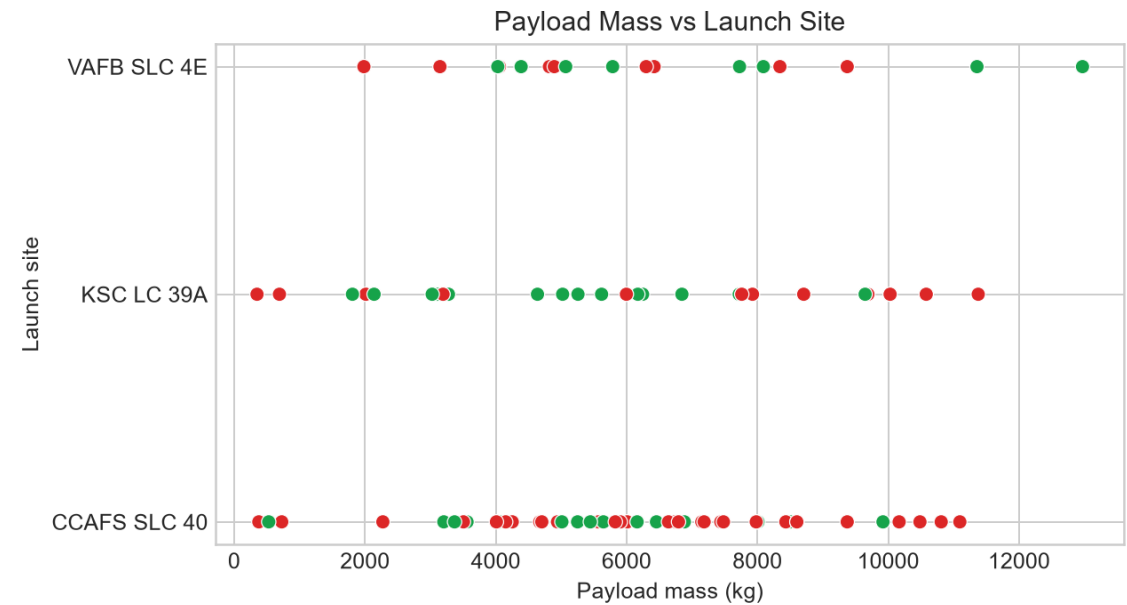


1.11 - EDA with Visualization Results - Scatter Plot 2

Rubric 1.11

Payload mass versus launch site

- Payload mass varies widely across sites and across landing outcomes.
- Payload alone does not explain success, so it should be interpreted with orbit, launch site, and flight history.
- This slide satisfies the required payload scatter plot result.

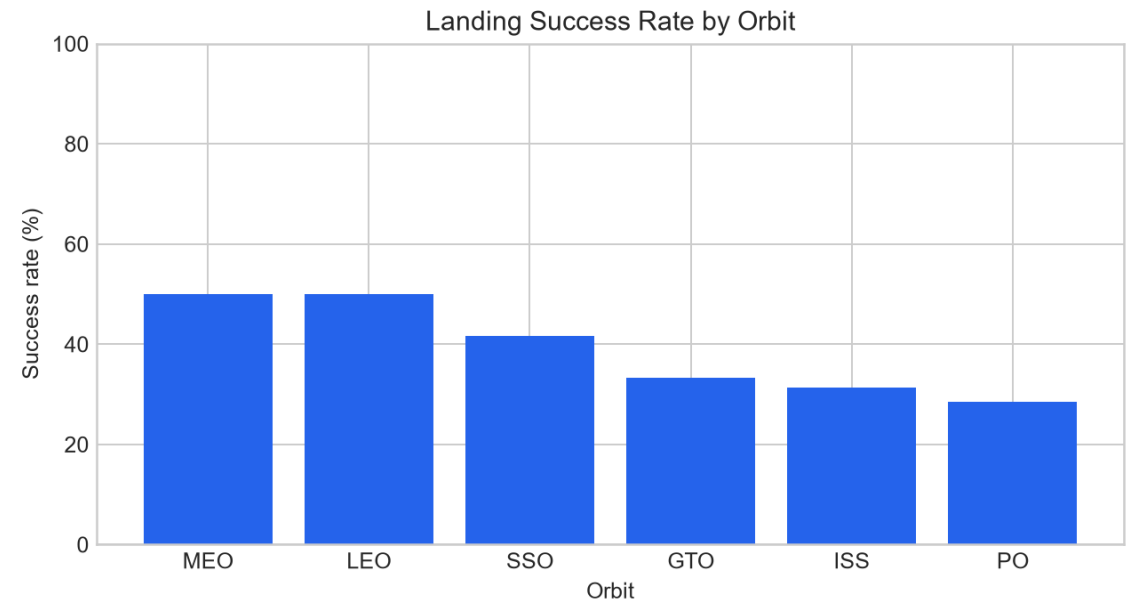


1.11 - EDA with Visualization Results - Bar Chart

Rubric 1.11

Landing success rate by orbit

- The bar chart shows that landing success differs by orbit.
- Orbit should be included in the model because mission profile affects landing outcome.
- This slide satisfies the required bar chart result.

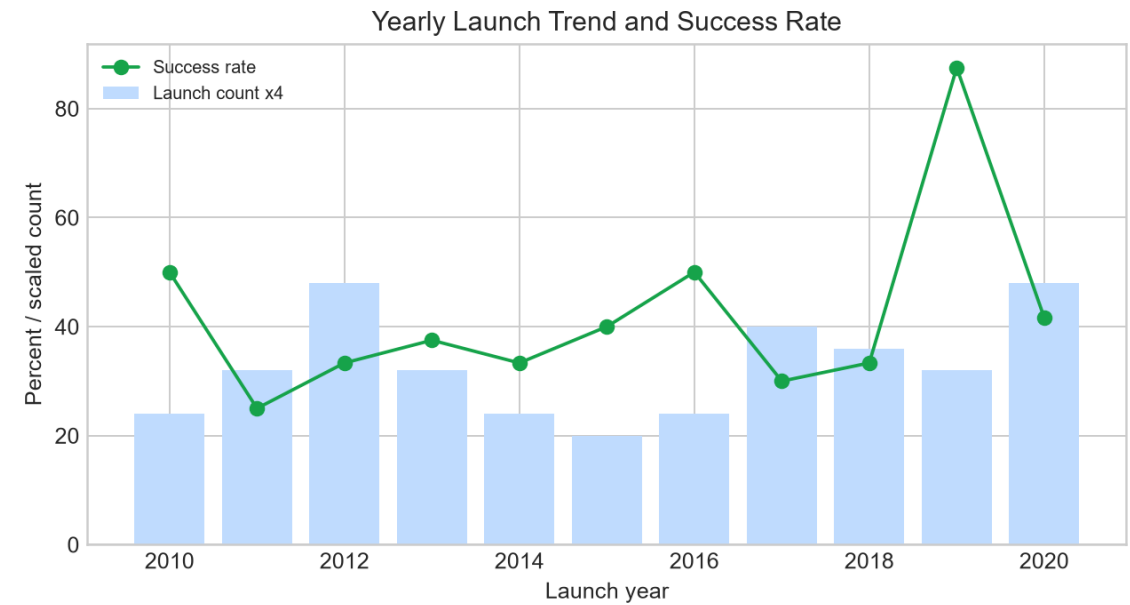


1.11 - EDA with Visualization Results - Yearly Trend

Rubric 1.11

Launch volume and success rate over time

- The yearly chart shows launch count and landing success rate across time.
- Landing success generally improves in later years, showing operational learning.
- This slide satisfies the required yearly trend result.



1.12 - EDA with SQL Results - Launch Sites, Success Rates, Rankings

Rubric 1.12

SQL result slide covering launch sites and rankings

SQL query: `SELECT LaunchSite, COUNT(*), AVG(Class), AVG(PayloadMass) FROM launches GROUP BY LaunchSite ORDER BY AVG(Class) DESC;`

Launch site	Launches	Success rate	Avg payload
CCAFS SLC 40	47	32%	6049 kg
KSC LC 39A	26	54%	5725 kg
VAFB SLC 4E	17	47%	6398 kg

SQL result: launch sites differ in launch volume, success rate, and average payload, so launch site is important for analysis and modeling.

1.12 - EDA with SQL Results - Payloads and Time Analysis

Rubric 1.12

SQL result slide covering payloads, success rates, and time analysis

Payload SQL query: `SELECT Class, MIN(PayloadMass), AVG(PayloadMass), MAX(PayloadMass), COUNT(*) FROM launches GROUP BY Class;`
Time SQL query: `SELECT LaunchYear, COUNT(*), AVG(Class) FROM launches GROUP BY LaunchYear ORDER BY LaunchYear;`

Outcome	Flights	Min payload	Avg payload	Max payload
Failure	53	350	6226	11376
Success	37	529	5729	12970

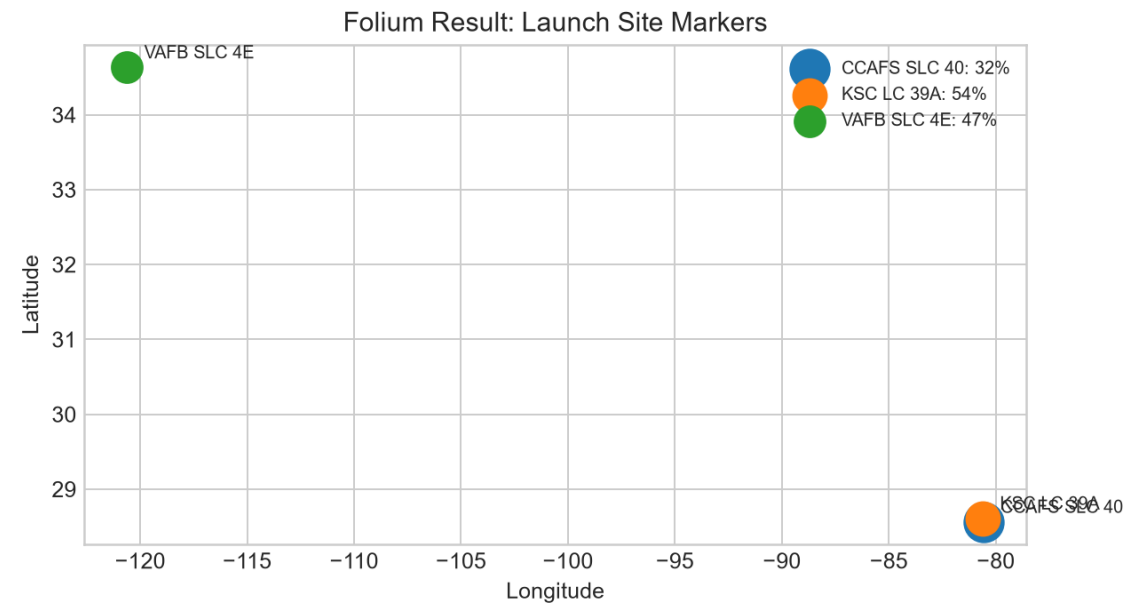
SQL result: payload and time patterns help explain success rates but are strongest when combined with site and orbit.

1.13 - Folium Map Results - Launch Site Markers

Rubric 1.13

Launch site marker map

- Each marker represents a launch site with latitude and longitude.
- Marker labels summarize launch site name and landing success rate.
- Florida sites are geographically close, while VAFB SLC 4E is on the west coast.

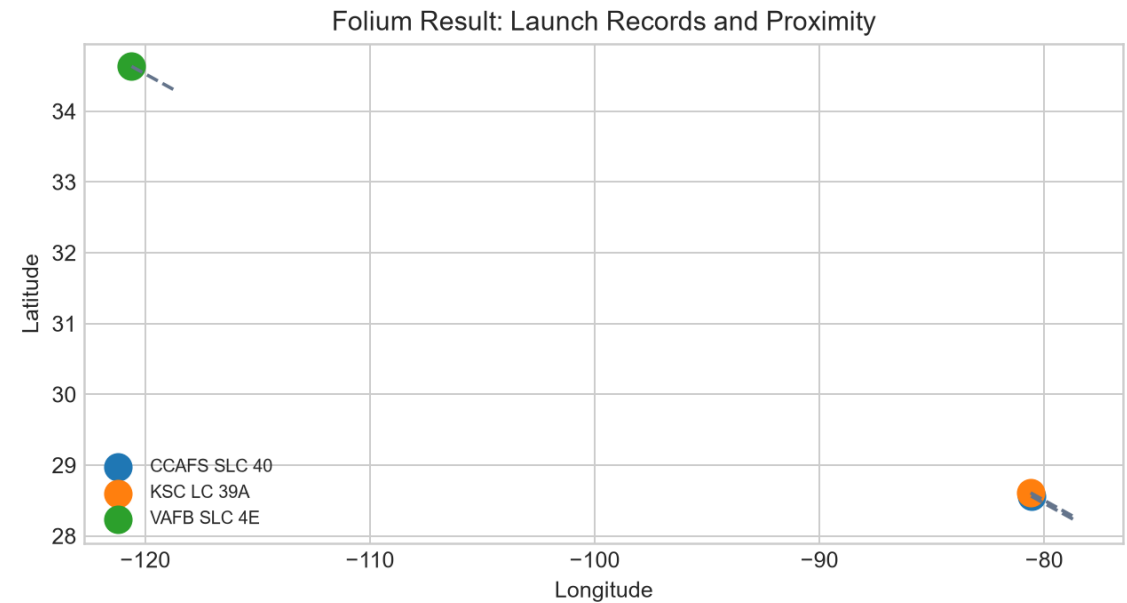


1.13 - Folium Map Results - Launch Records and Proximity Analysis

Rubric 1.13

Launch records and proximity map

- Launch records were mapped with outcome markers to compare successes and failures.
- Proximity analysis measured distances to coastlines, cities, highways, and railways.
- Operational insight: launch-site geography supports safety, accessibility, and mission trajectory planning.

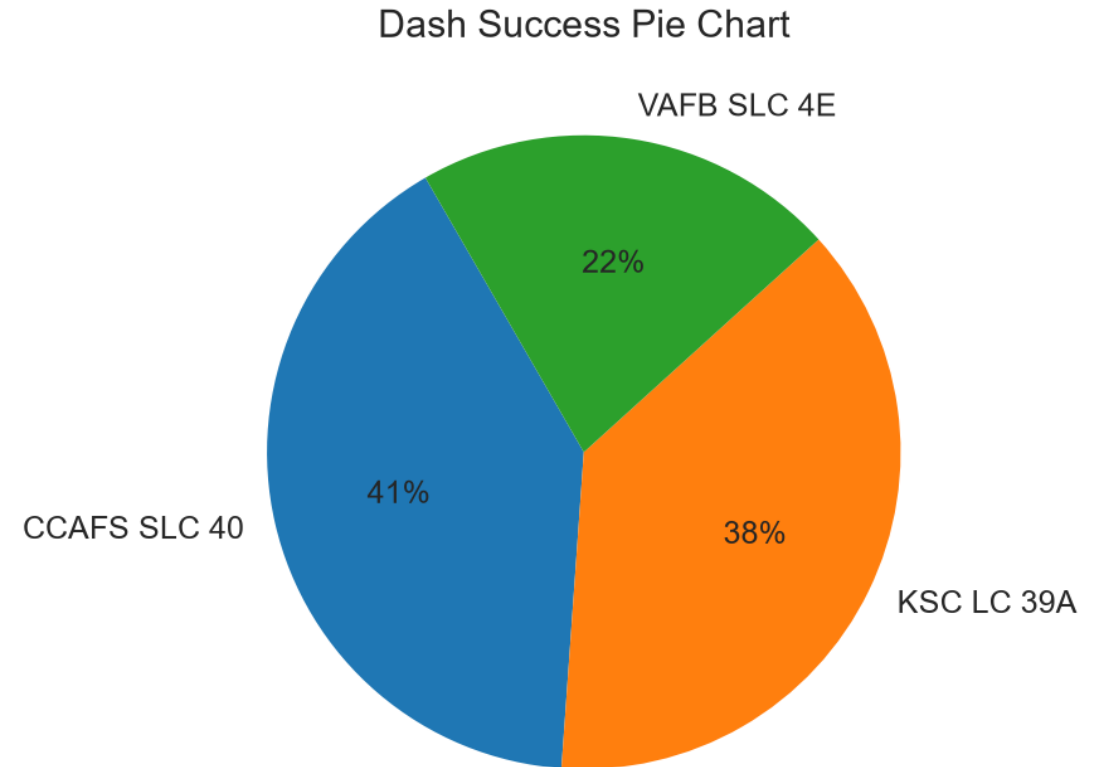


1.14 - Plotly Dash Results - Success Pie Chart

Dashboard success pie chart

Rubric 1.14

- The dashboard includes the title SpaceX Launch Records Dashboard.
- The site dropdown can show all launch sites or one selected site.
- The success pie chart summarizes successful launches by site or selection.



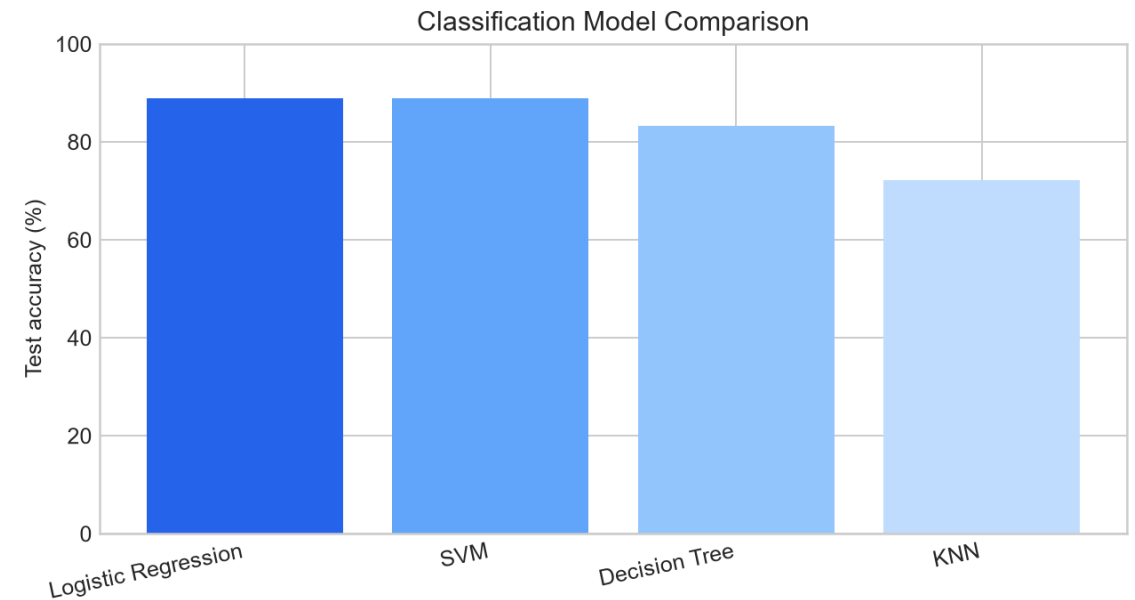
Rubric 1.14

1.15 - Predictive Analysis Results - Multiple Models and Evaluation

Rubric 1.15

Model comparison and evaluation results

- Four models were compared: Logistic Regression, SVM, Decision Tree, and KNN.
- Logistic Regression and SVM reached the strongest test accuracy at 88.9%.
- Logistic Regression was selected as the best model because it matched the top accuracy and remained easier to interpret.

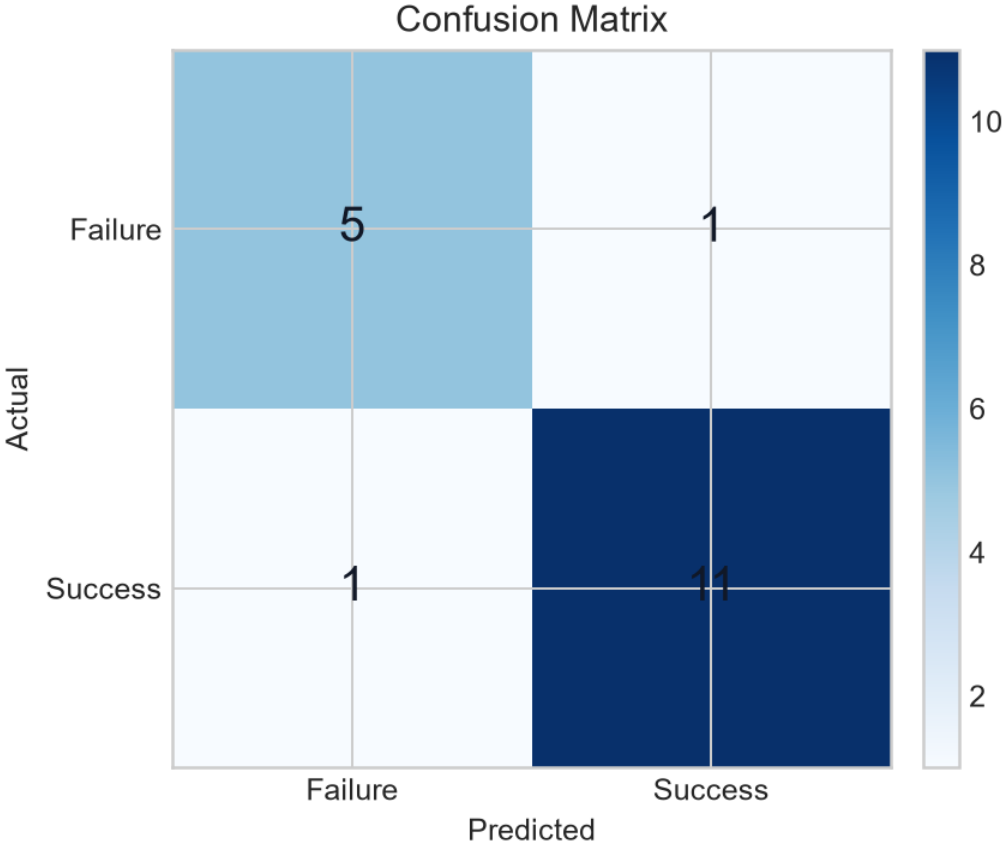


1.15 - Predictive Analysis Results - Confusion Matrix and Best Model

Rubric 1.15

Confusion matrix and model behavior

- The confusion matrix compares actual and predicted landing outcomes.
- Most test records were classified correctly, which shows useful signal in the prepared features.
- False predictions still matter because landing failure affects cost, reuse planning, and bid risk.



1.15 - Conclusion

Rubric 1.15

Findings, recommendation, and model refinement

- Main finding 1: Falcon 9 landing success is connected to launch site, orbit, payload mass, flight history, and booster features.
- Main finding 2: EDA, SQL, Folium, Dash, and classification results all show that mission context matters more than any single variable.
- Best model conclusion: Logistic Regression was selected from multiple models because it reached top test accuracy and stayed interpretable.
- Recommendation: update the project with newer SpaceX records, add weather and mission complexity features, then retrain and compare models again.
- Real-world use: the model can support early launch-bid screening by estimating the likelihood of first-stage recovery.

1.15 - Creativity and Innovative Insights

Rubric 1.15

Creative interpretation beyond the required charts

- Creative element: the presentation connects the full workflow from raw data collection to a business decision about launch bidding risk.
- Innovative insight 1: flight number and launch year act as operational-learning indicators, not only record identifiers.
- Innovative insight 2: nearby launch sites can still produce different outcome patterns, so geography should be interpreted with site-specific operations.
- Innovative insight 3: payload mass is most useful when combined with orbit and site; isolated payload charts can hide mission-profile effects.
- Business innovation: the dashboard and model can become a simple recovery-risk screening tool before pricing a launch proposal.